28TH INTERNATIONAL KARSTOLOGICAL SCHOOL "CLASSICAL KARST"
REGIONAL KARSTOLOGY - LOCAL AND GENERAL ASPECTS

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MAP OF KARST PHENOMENA OF NATIVIDADE AND CHAPADA DA NATIVIDADE, TOCANTINS, BRAZIL

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INTRODUCTION

Given the lack of geomorphological research in the Region of Nativity and Chapada da Nativity, Tocantins, and the need to understand the evolution of the karst landscape on carbonate rocks, the research aims to identify the different relief compartments, with emphasis on the main karst features distributed among them.

The authors also made onsiderations about the stage of landscape conservation, and the mains features were placed, in an exploratory map of karst phenomena, on a semi-detail scale (1:50,000).

Finally, the work aimed to demonstrate the tourist potential of the studied area.

METHODOLOGICAL PROCEDURES AND MATERIALS

1

- **Cartographic Bases**
  - Satellite and Radar images
    - Sentinel 2A
      Alos/Palsar
      Topographic maps
    - United States Geological Survey
      Alaska Satellite Facility
      Brazilian Army Geographical Service
  - Vector Data
    - Streams
      Roads
      Municipal Limits
    - SEPLAN – TO
      IBGE

2

- **Bibliography**
  - Karst
    - Tricart (1956)
      Sallun Filho e Karmann (2007)
      Heidari et al. (2011)
      Sauro (2012)
      Kranjc (2013)
      etc.
    - Brazilian Geographic Review
      Jornal of Maps
      Geomorphology
      Encyclopedia of Caves
      Treatise on Geomorphology
  - Landscape
    - Vitte (2007)
      Leite (2011)
      Cavalcanti (2014)
      Ab'Saber (2012)
      etc.
    - Mercator
      Doctoral Thesis
      Landscape Cartography
      Domains Nature of Brazil

1 + 2 + FIELD WORK

→ Delimitation of gemorphological compartments

→ MAP OF KARST PHENOMENA

RESULTS

Three specific compartments were delimited:

1) The Non-carbonatic metasedimentary plateau (PMNC)
2) The Carbonate Plateau (CPC), and
3) The Surface with occurrence of dolinas (SCOD)

In this context, the main karst features are distributed in the CPC and SCOD. The researched area stands out for presenting structurally controlled karst hills, especially the hills of Oficina, Carneiro, Araras, Mutum and Serra Preta. The depressions correspond to cover colapse dolines, ponors, and a contact polje described based on Nicod (2003), Heidari et al. (2011), Sauro (2012), Kranjc (2013). These features were synthesized in Figure 1.

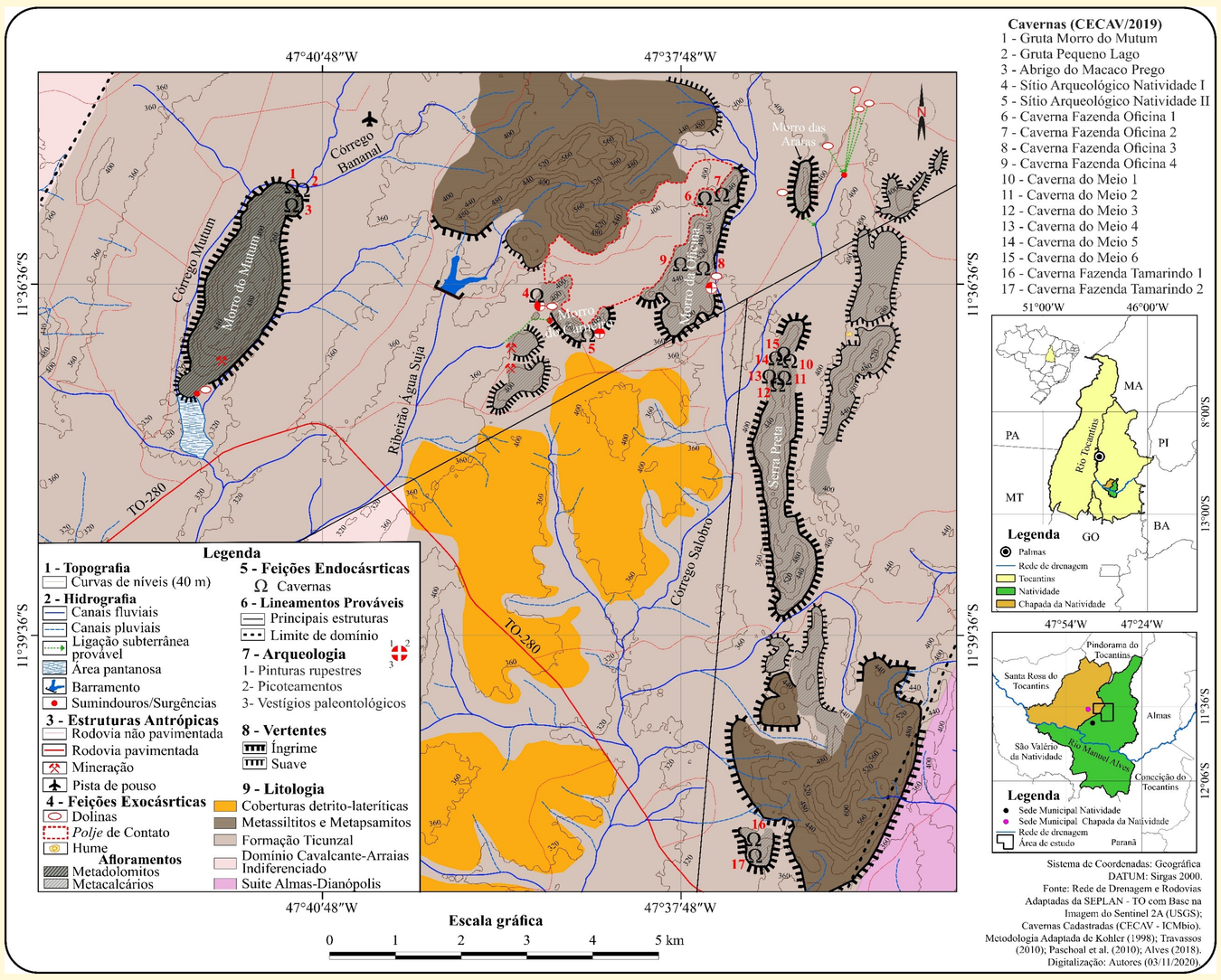


Fig. 1: Map of karst phenomena of Natividade and Chapada da Natividade, Tocantins, Brazil

CONCLUSIONS

In the area, a well-developed traditional karst relief was recognized in carbonate rocks. Moreover, the work showed the scientific and geotourism potential of the region, emphasizing the need for further research, which aims to understand the regional karst better and enable its responsible use.

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